Aaditya Bhat - EECS10B Extra Credit #4

1) Check if R16 decremented to 0 (5 points)

```
DEC R16
CPI R16, 0
BREQ LastRow
```

```
DEC   R16
BREQ  LastRow
```

2) Check if R16 decremented to 0xFF (5 points)

```
DEC R16
CPI R16, 0xFF
BREQ NextRow
```

```
SUBI  R16,0x01 ;dec but with carry flag set
BRCS  NextRow  ;only sets carry if 0->FF on dec
```

3) Sgn function of R16 (12 points)

Given:	`R16` - 8-bit signed value
Result:	`R16` - sgn function of the original value in R16 (-1 for R16 < 0, 0 for R16 = 0, and +1 for R16 > 0)
Code:	can be done in 5 instructions

```
LSL R16 ;r16=2*r16, c=msb(r16)

SBC R17,R17; R17 = 0xFF if negative, 0x0 if positive

CP  R17,R16

CLR R16

ADC R16,R17 ; If neg:0+(-1)+0=-1; If zero:0+0+0=0; If pos:0+0+1=+1
```

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4) Increment time value in `R17|R16` (8 points)

Given: `R17|R16` - 16-bit time value in BCD, R17 is the minutes (0 to 99) and R16 is the seconds (0 to 59)

Result: `R17|R16` - the original time incremented by 1 second in the same format

```
IncSeconds:
 SUBI R16,0xFF          ;INC
 MOV  R18,R16
 ANDI R18, 0x0F
 CPI  R18, 0x0A        ;are we bigger than 10?
 BRNE CheckSecMax
 SUBI R16,0xFA         ;add 6 to correct BCD
CheckSecMax
  CPI  R16, 0x60       ;reached 60?
  BRNE Done
  CLR  R16
  SUBI R17, 0xFF       ; add 1 to mins
  MOV R18, R17
  ANDI R18, 0x0F       ; check nibble of mins
  CPI R18, 0x0A
  BRNE CheckMinMax
  SUBI R17, 0xFA       ; add 6 for BCD correction
CheckMinMax
  CPI R17, 0x60
  BRNE Done
  CLR R17
Done ;Execution finished
```